Distributed Systems. Introduction

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1 Introduction

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1.2 Properties of distributed systems

1.3 Distributed applications

1.4 Hardware support

1.5 Software support

1.6 Structure of a distributed system

1.1 The road to distributed systems

- Objective:
  - resource (service/device) sharing
- Types of systems (historical evolution):
  - batch systems: deferred processing, sequential
  - time sharing centralized systems: terminals
  - teleprocessing systems: telephone network
  - personal systems: workstations, PCs
  - networked systems: client/server, protocols (TCP/IP)
  - distributed systems: transparency (GUI, RPC/RMI)

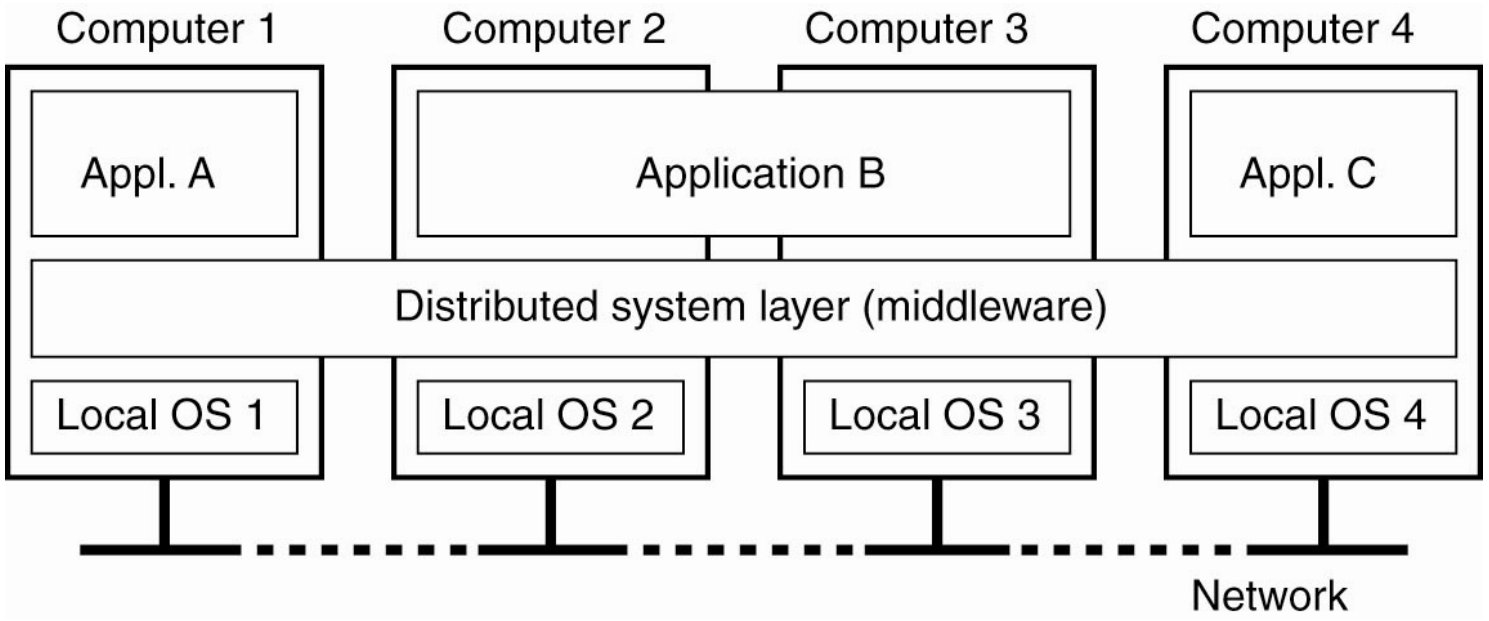
1.1 The road to distributed systems

"A distributed system is a collection of independent computers that appears to its users as a single coherent system"

- A definition of distributed system
  - (1) collection of *computers*
  - (2) interconnected
    - as a networked system
  - (3) shared state
  - (4) single system image (*SSI*)
    - as a centralized system

1.1 The road to distributed systems

- A distributed system is software ⇒ *Middleware*



A distributed system organized as middleware. The middleware layer extends over multiple machines, and offers each application the same interface

1.1 The road to distributed systems

- Advantages with respect to centralized systems:
  - low cost: can be composed of standard PCs
  - scalability: consequence of modularity
  - flexibility: reuse of old machines
  - availability: resource replication
  - allow parallel processing
  - a way of accessing remote resources
- Advantages with respect to networked systems:
  - efficient resource usage: by means of migration
  - transparent access to resources with respect to location

1.1 The road to distributed systems

- Drawbacks with respect to centralized systems:
  - a centralized system of the same cost is more efficient than any computer of the distributed system
    - important for sequential applications (e.g., computation)
  - due to the cost of communications, if the physical location of resources is inadequate, some may be flooded while others are free
  - maintaining consistency can be very expensive
  - the network can be a source of problems
  - security management is more complex

1.1 The road to distributed systems

- Current trend:
  - Mobile computing
    - Laptops, smartphones…
    - Wireless networks
  - Ubiquitous/pervasive systems
    - Computers/sensors everywhere: houses (domotics), cars, buildings (e.g., hospitals)
    - Pervasive environments are very dynamic
    - Protocols for resource discovery: Jini, UPnP…

1.2 Properties of distributed systems

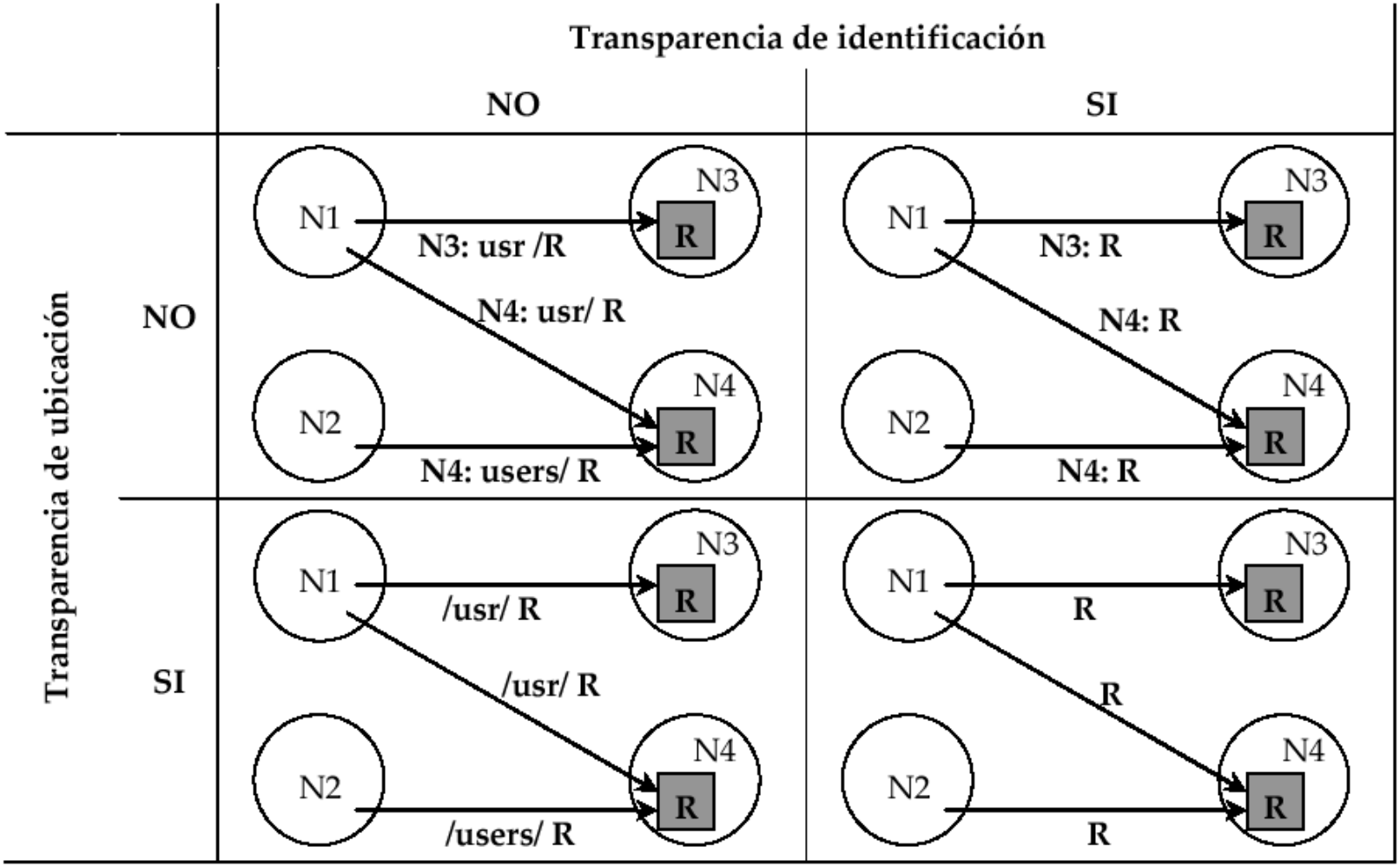
- Objective:
  - Single System Image
- Desired properties:
  - Transparency
  - Scalability
  - Dependability
  - Consistency

Transparency in a Distributed System

⌘**Transparency**	⌘**Description**
⌘Access	⌘Hide differences in data representation and how a resource is accessed
⌘Location	⌘Hide where a resource is located
⌘Migration	⌘Hide that a resource may move to another location
⌘Relocation	⌘Hide that a resource may be moved to another location while in use
⌘Replication	⌘Hide that a resource is replicated
⌘Concurrency	⌘Hide that a resource may be shared by several competitive users
⌘Failure	⌘Hide the failure and recovery of a resource
⌘Persistence	⌘Hide whether a (software) resource is in memory or on disk

Different forms of transparency in a distributed system (ISO, 1995)

Transparency in a Distributed System



Access and location transparency

1.2 Properties of distributed systems

- Scalability
  - Grow capacity, without damaging efficiency
  - Based on modularity
  - Namespaces:
    - used for identifying objects of different nature: files, processes, variables, memory addresses
    - linear namespaces (memory): 32 bits is not enough
    - generally hierarchical, and therefore scalable
  - Maintaining efficiency: replication
    - mirroring, caching
    - achieving replication transparency is complex

1.2 Properties of distributed systems

"A distributed system is a system in which the crash of a computer you've never heard of stops you from getting any work done" (Leslie Lamport)

- Dependability

  "Ability of a system at any time to perform its work as expected"

  - Availability:
    - Time in which the system works correctly (%)
      - parameters: MTBF (Mean Time Between Failures), MTTR…
      - improving the quality of components vs replication (cheaper)
  - Fault-tolerance:
    - Ability to maintain a correct functioning despite the failure of a component
      - replication (passive, active)

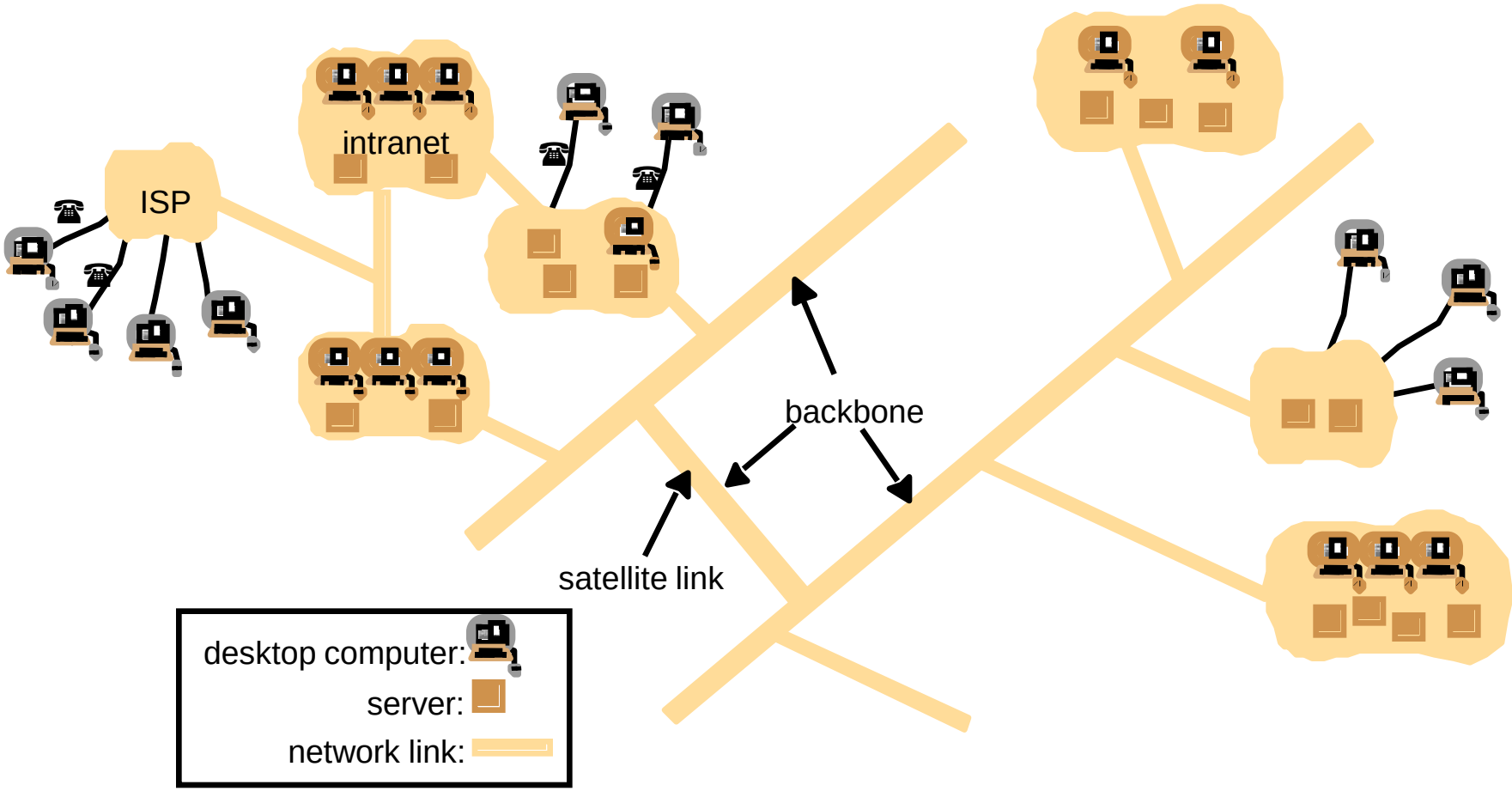
1.2 Properties of distributed systems

- Consistency
  - Problems related to replication:
    - copy interconnection is a source of error
    - confidentiality is more difficult to maintain
    - management of global state is complicated
  - Problems to maintain consistency:
    - physical distribution: several copies, each with its own state
    - communication errors and/or delays
    - no global clock: how to order events?
  - Transactions (ACID), group communication…
  - Improving efficiency: relax consistency

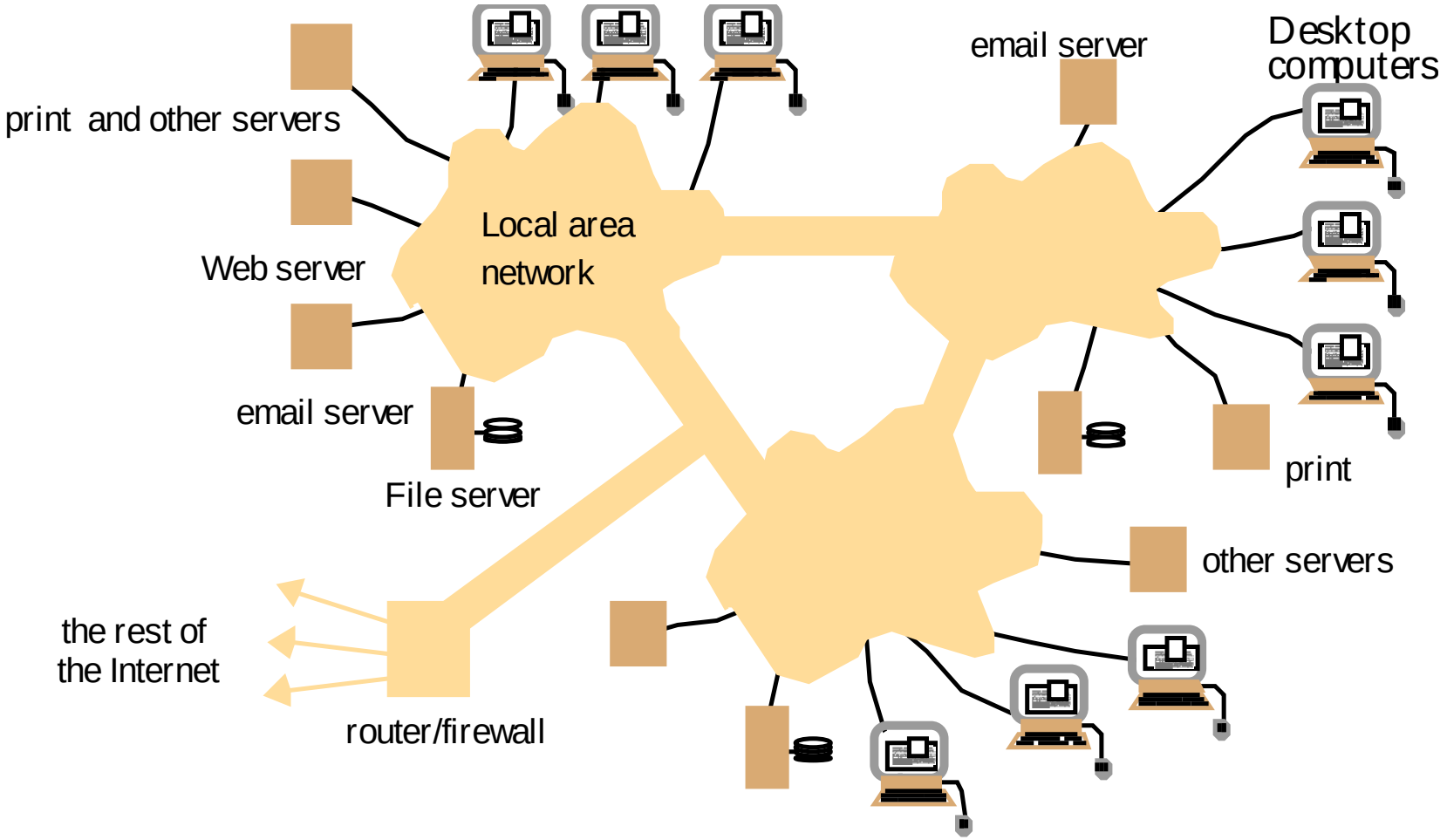
1.3 Distributed applications

- Parallel applications: many tasks at a time
  - main goal: reduce execution time
- Distributed applications (motivation):
  - high performance: cluster computing
  - fault-tolerance: replication, transactions
    - banking computer systems
    - consistency is critical
  - high availability: caching, mirroring
    - inherent distribution: Internet (Web), file-systems
    - consistency is important, but not critical
  - mobility, ubiquity: AmI applications

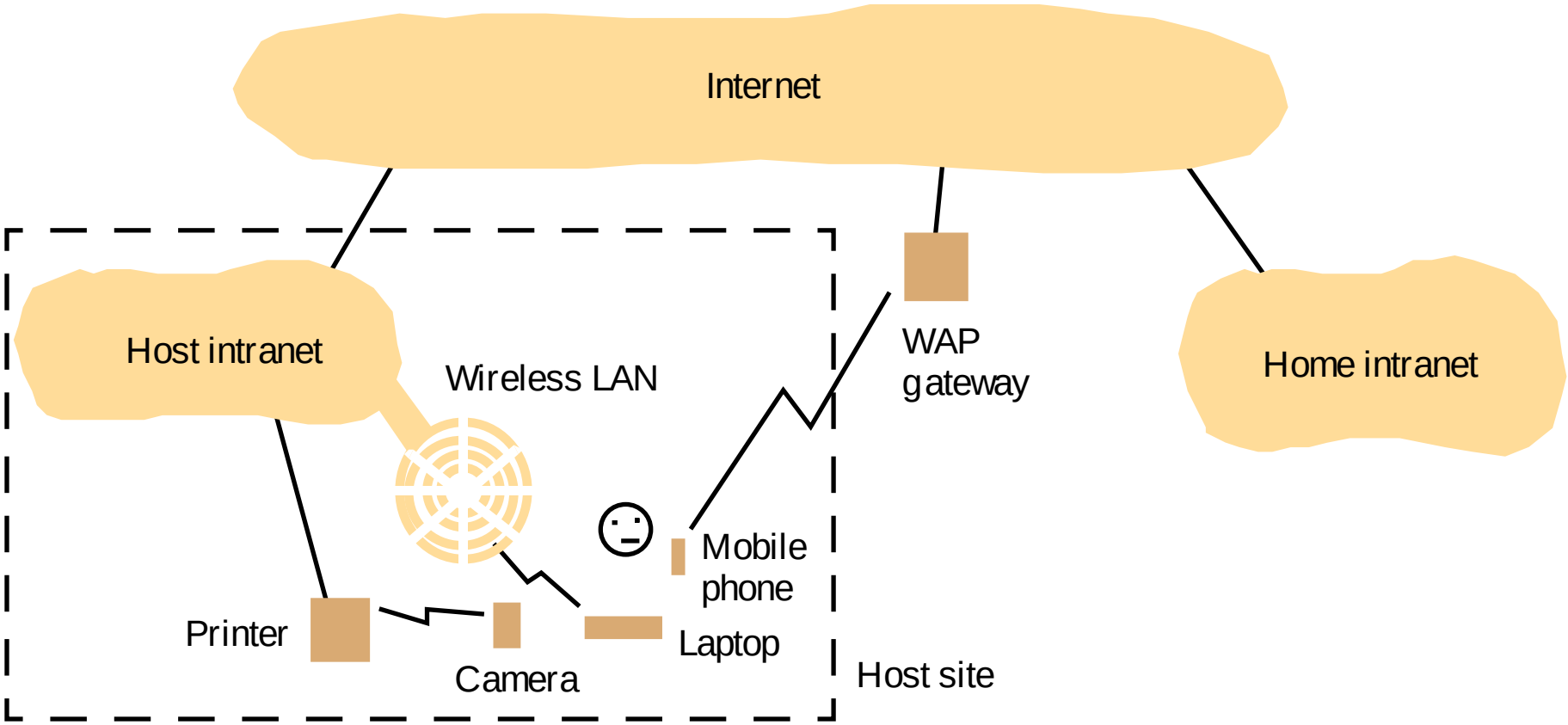
The Internet as a Distributed System



An intranet as a Distributed System



Mobile/Ubiquitous Computing as a Distributed System



Some Internet related numbers

Date	*Computers*	*Web servers*	*Percentage*
1979 December	188	0	%0
1989 July	130,000	0	%0
1993 July	1,776,000	130	%0.007
1995 July	6,642,000	23,500	%0.4
1997 July	19,540,000	1,203,096	%6
1999 July	56,218,000	6,598,697	%12
2001 July	126,000,000	30,000,000	%24
2004 July	285,000,000	68,000,000	%24
2007 July	490,000,000	125,000,000	%25

1.4 Hardware support

- What is a computer?

data / instruction	**single data at a time**	**multiple data at a time**
single instruction at a time	*SISD* Von Neumann traditional architecture	*SIMD* vector processors
multiple instructions at a time	*MISD* not implemented	*MIMD* multiprocessors, multicomputers, networks

Flynn's classification

1.4 Hardware support

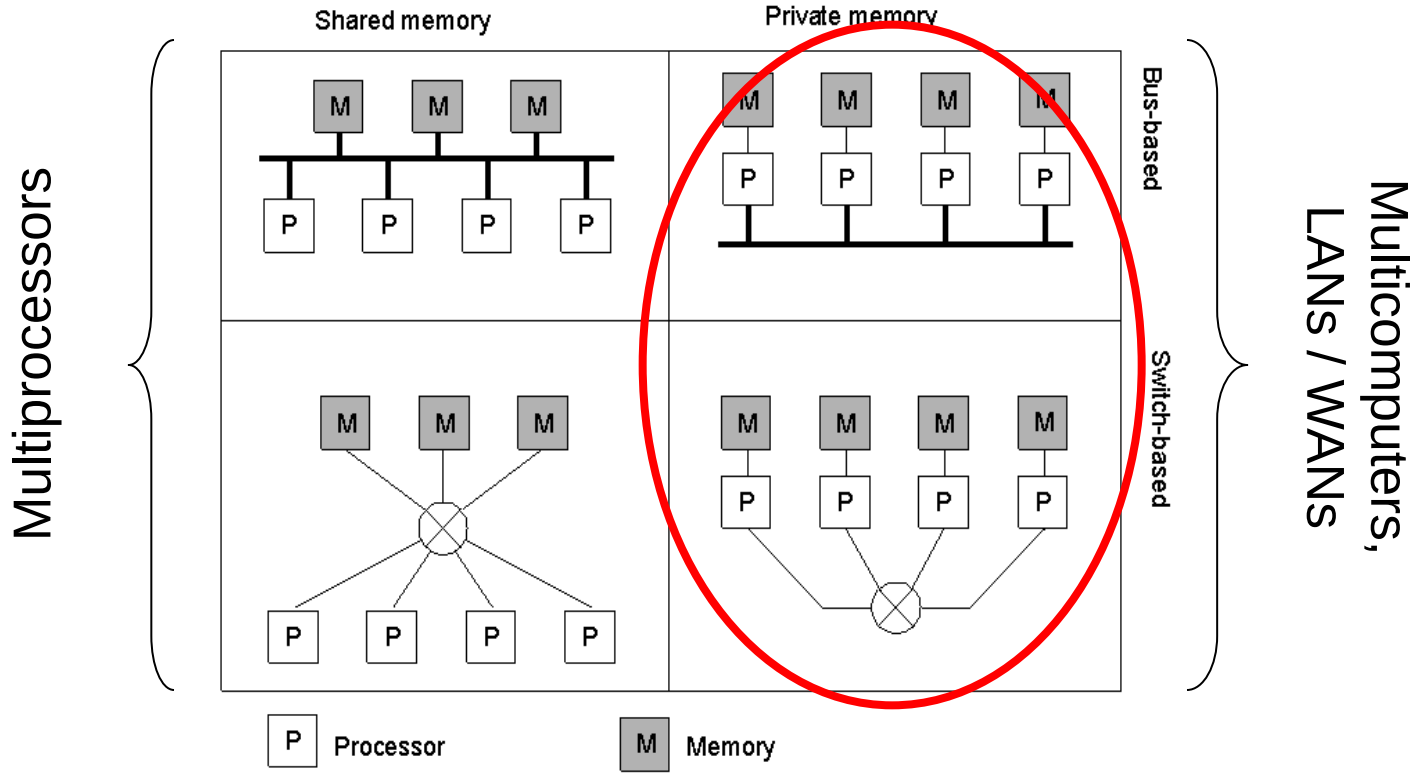
- *MIMD*: coupling-degree and inter-connection

coupling-degree / inter-connection	**physical shared memory**	**independent memory allocations**
shared bus	multiprocessors	multicomputers, *LANs*
interconnection network	*UMA* and *NUMA* multiprocessors	multicomputers, *WANs* (*Internet)*

MIMD architectural types

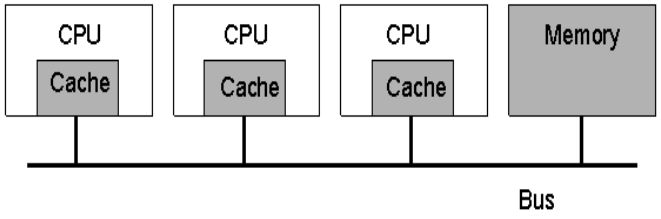
1.4 Hardware support

- *MIMD*: coupling-degree and inter-connection

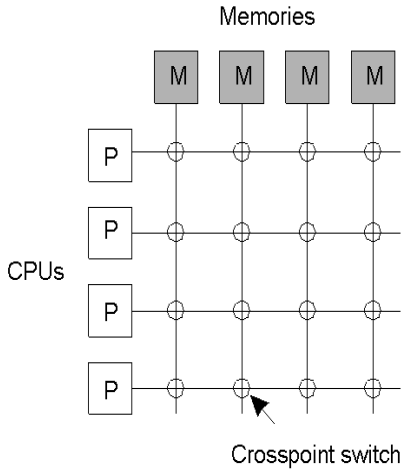


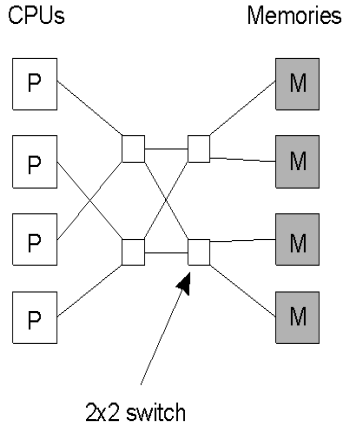
1.4 Hardware support

- Multiprocessors



Bus-based multiprocessor





a) Crossbar switch

b) Omega switching network

1.4 Hardware support

- Multicomputers

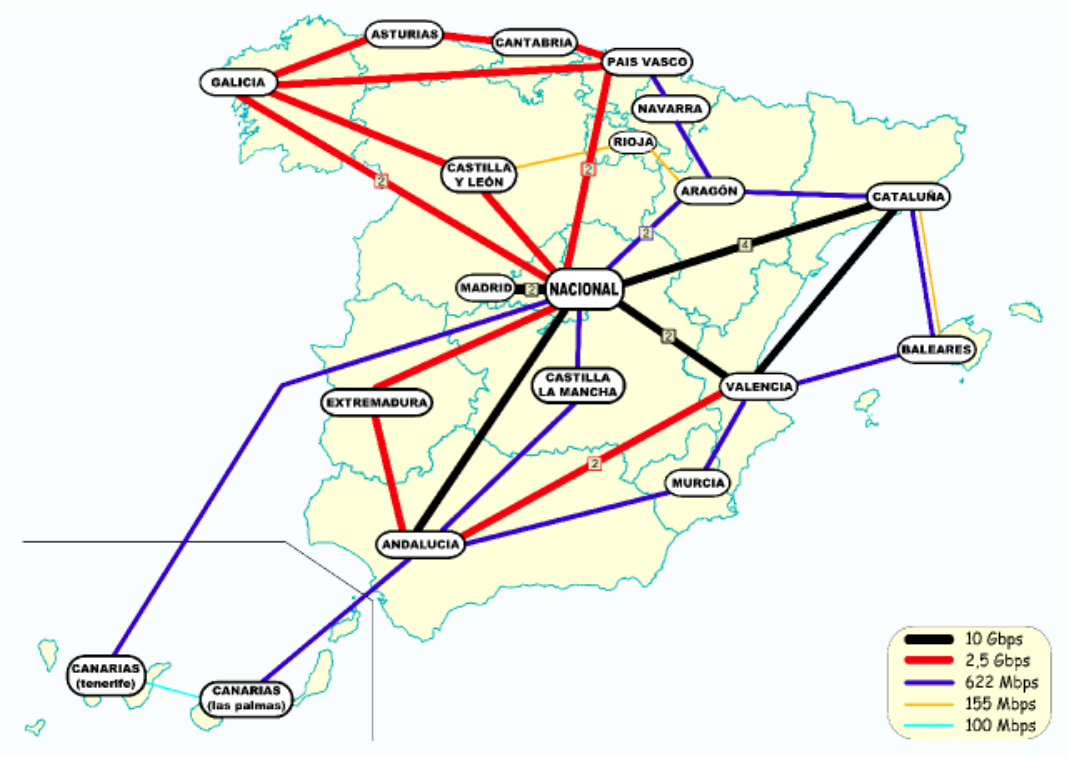




1.4 Hardware support

- LANs / WANs





RedIRIS network

1.4 Hardware support

- Communication networks:
  - Wired:
    - PAN: USB (1 m, 12 - 480 Mbps)
    - LAN: Ethernet (1 km, 10 - 1000 Mbps)
    - MAN: ATM (10 km, 1 - 150 Mbps)
    - WAN: Internet (Worldwide, 0,5 - 600 Mbps)
  - Wireless:
    - PAN: Bluetooth (10 m, 0,5 - 2 Mbps), Zigbee, IrDA
    - LAN: WiFi (100 m, 2 - 54 Mbps)
    - MAN: WiMAX (10 km, 1,5 - 20 Mbps)
    - WAN: UMTS-HSDPA (Worldwide, up to 14 Mbps)

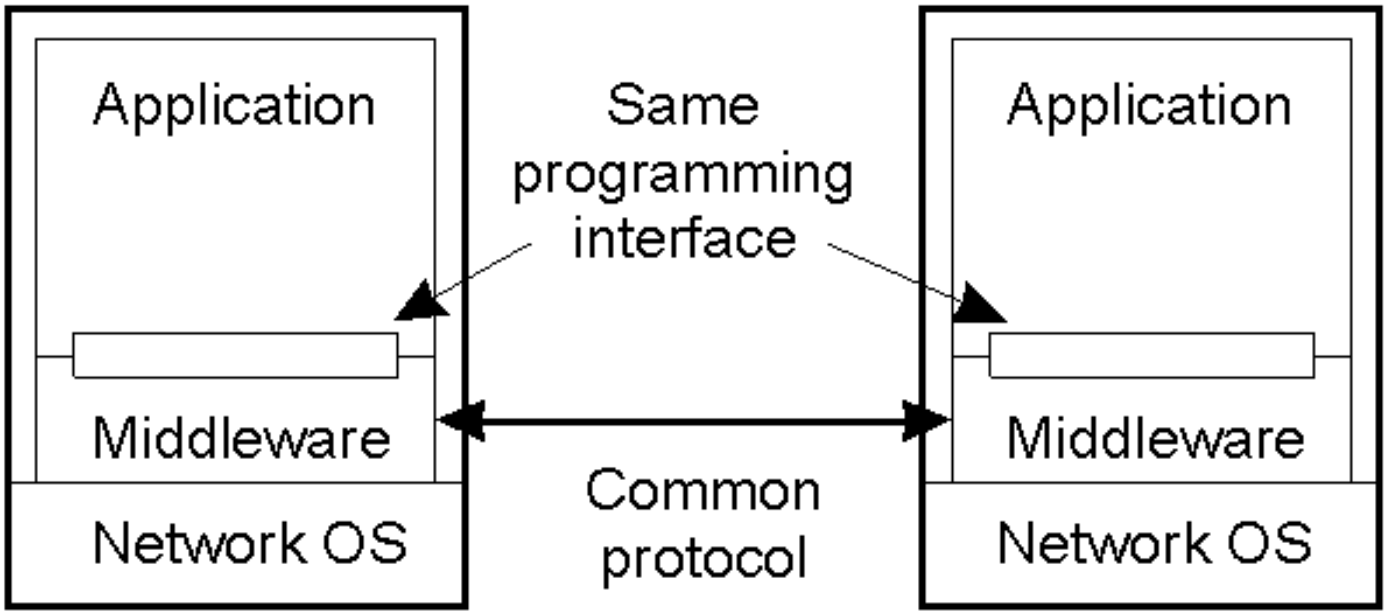
1.5 Software support

- Hardware support of distributed systems:
  - set of nodes, each one with its own memory, I/O, operating system and basic network services
  - Examples: multicomputers, LANs / WANs
- Integration problem: heterogeneity
  - hardware, operating system…
- Solution: open systems
  - public interface specifications
  - standards: officials vs *de facto* (OSI vs TCP/IP)

1.5 Software support

- Properties of open systems:
  - Interoperability
    - Thanks to standard protocols: TCP/IP, RPC/XDR
    - Interface Definition Languages: CORBA IDL
      - Trend: XML/SOAP (Web services)
  - Application transportability
    - POSIX (source code, from one Unix to another)
    - Java (executable code, from one JVM to another)
  - User transportability: GUI, NIS

Middleware and Openness



- In an open middleware-based distributed system, the protocols used by each middleware layer should be the same, as well as the interfaces they offer to applications.

1.5 Software support

- Communication support:
  - Physical shared memory systems:
    - shared variables (synchronization functions, *Test&Set*)
  - Physical distributed memory systems:
    - message passing (network protocols)
  - The physical organization of memory (coupling-degree) and the communication model must be distinguished:
    - the communication model may be shared memory or message passing
    - the coupling-degree determines the communication model support (but not the model itself!)

1.5 Software support

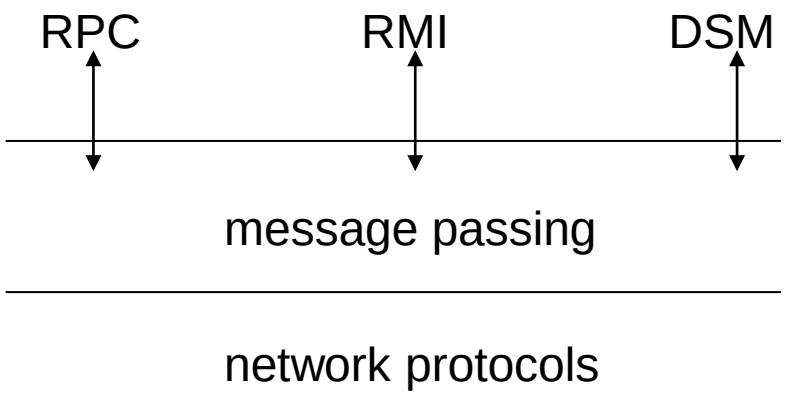
- Coupling degree and communication model:

Coupling degree / Communication model	Physical shared memory	Physical distributed memory
Shared memory	Shared variables	Distributed Shared Memory (DSM), distributed objects (RMI)
Message passing	pipes, FIFOs, UNIX sockets	INET sockets, MPI, RPC

Communication mechanisms

1.5 Software support

- Implementing communication mechanisms:
  - Basic standard message passing: INET sockets
    - client/server model
    - blocking / non blocking
    - reliable (TCP) / unreliable (UDP)
    - point-to-point / broadcast / multicast (IP Multicast)
  - RPC, RMI, DSM:
    - based on message passing



1.5 Software support

- Operating system support:
  - Desired properties: open and flexible
    - efficient development, location and management of services
  - Classic OS (UNIX) are monolithic, rigid: all services in the kernel, single set of system calls, pre-defined management politics
  - Alternatives and trends:
    - Hw emulation (VMware, Virtual PC): many OS at a time
    - Microkernels (Mach): most services out of the kernel
    - Smartphones: adapted OS (iOS, Windows Mobile, Symbian, Android) + Web browser (HTTP)

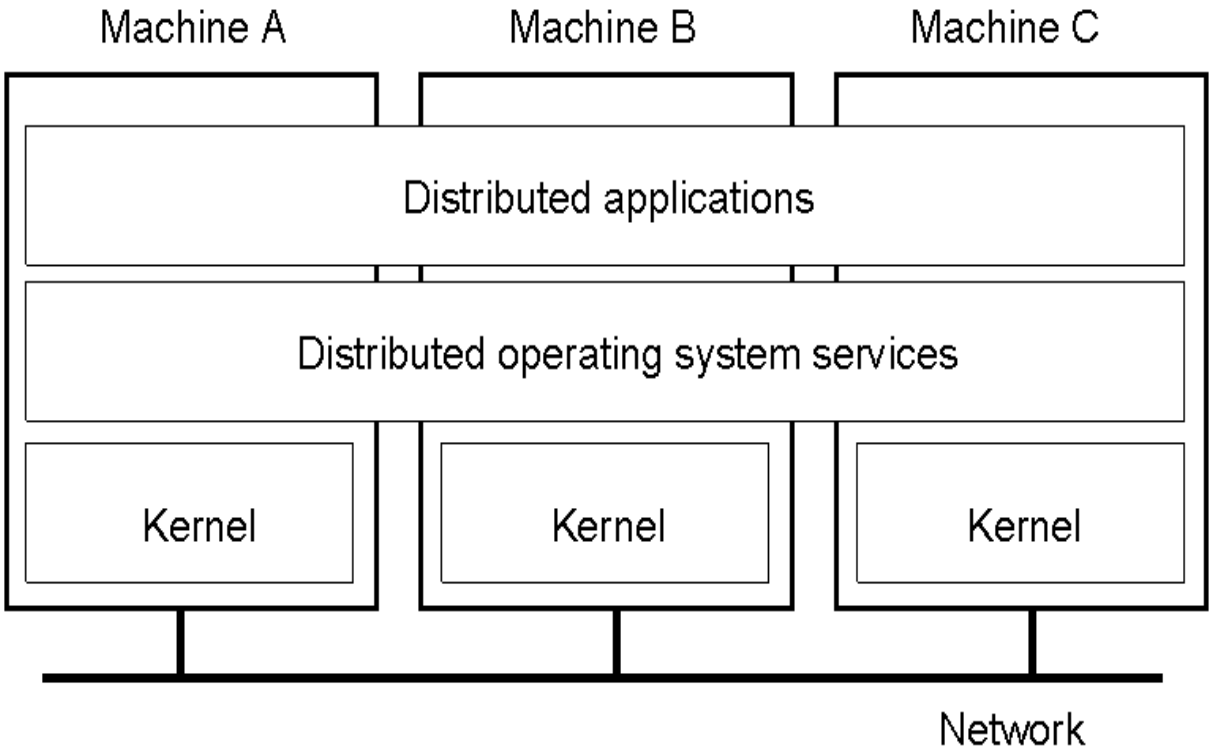
1.5 Software support

- Distributed Operating Systems (DOS), Network Operating Systems (NOS) and Middleware

System	Description	Main Goal
DOS	Tightly-coupled operating system for multi-processors and homogeneous multicomputers	Hide and manage hardware resources
NOS	Loosely-coupled operating system for heterogeneous multicomputers (LAN and WAN)	Offer local services to remote clients
Middleware	Additional layer atop of NOS implementing general-purpose services	Provide distribution transparency

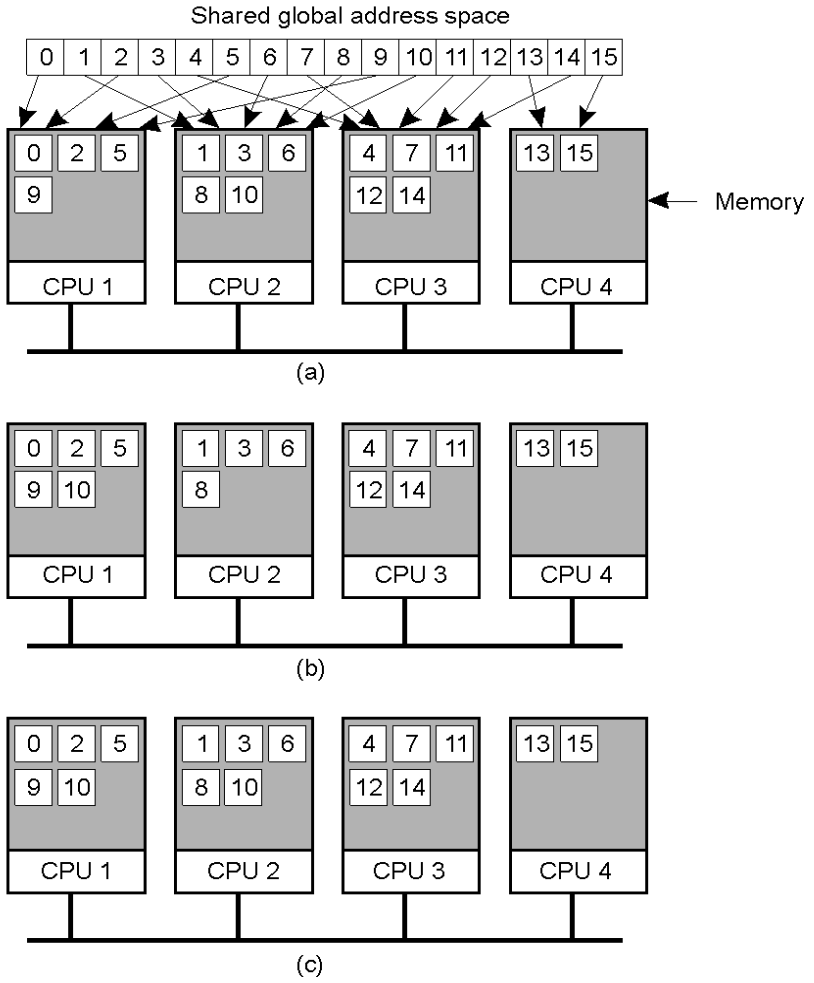
1.5 Software support

- Distributed Operating System (DOS) in a multicomputer



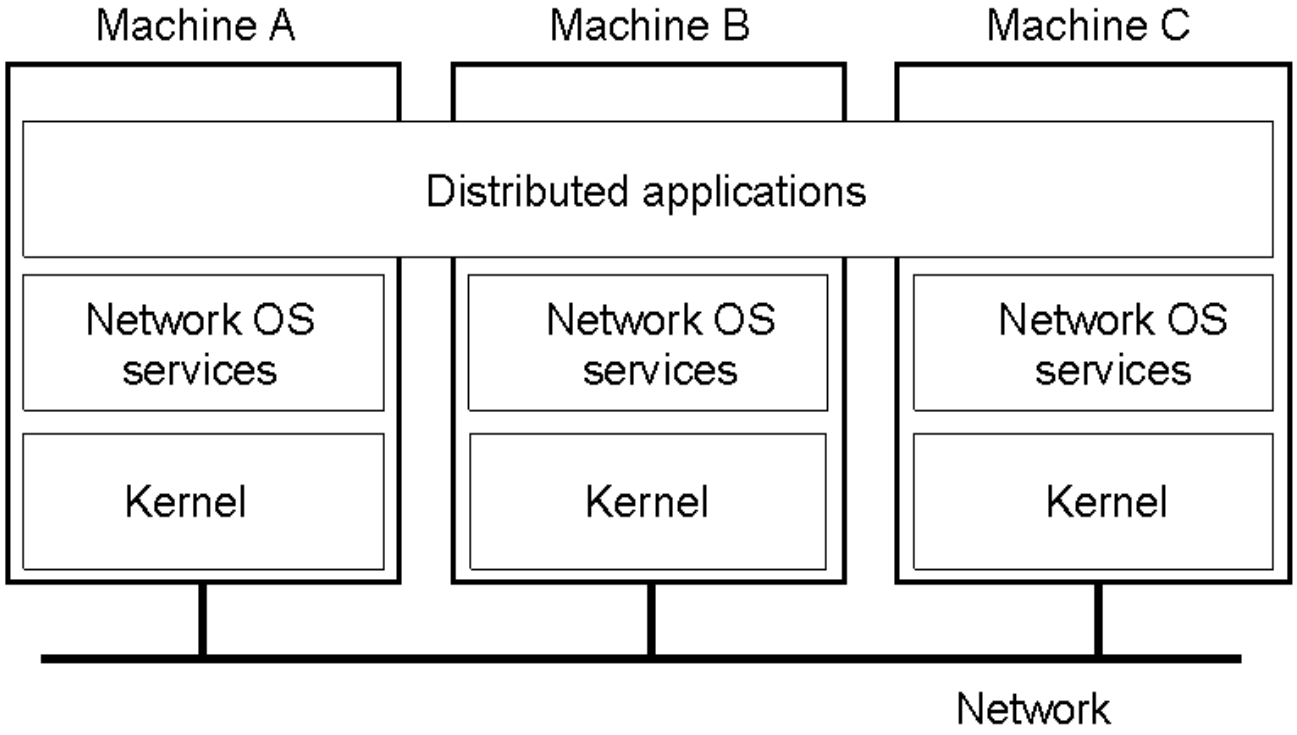
1.5 Software support

- DOS service: Distributed Shared Memory
  - a) Pages of address space distributed among four machines
  - b) Situation after CPU 1 references page 10
  - c) Situation if page 10 is read only and replication is used



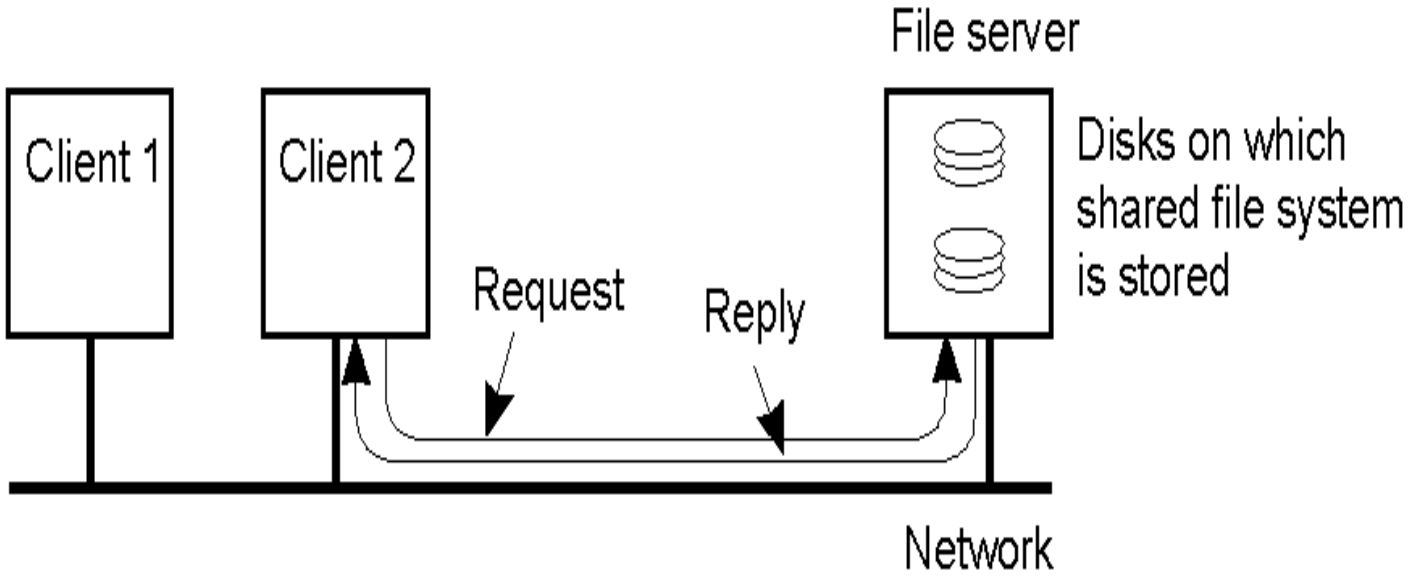
1.5 Software support

- Network Operating System (NOS)



1.5 Software support

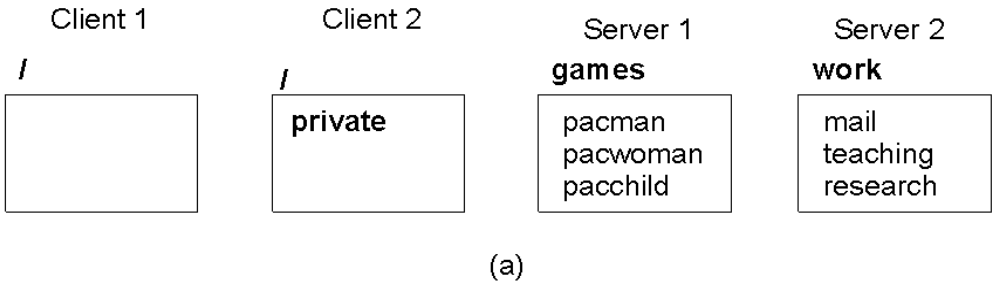
- Network Operating System (NOS)



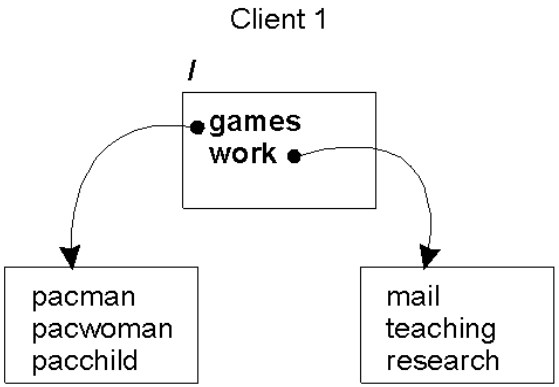
Two clients and a server in a NOS (e.g., *NFS*)

1.5 Software support

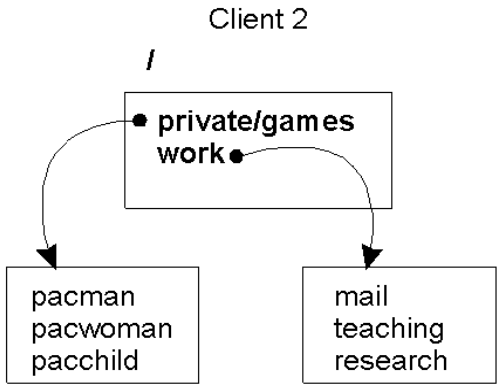
- Network Operating System (NOS)
  - Clients can mount the server at different points



(a)



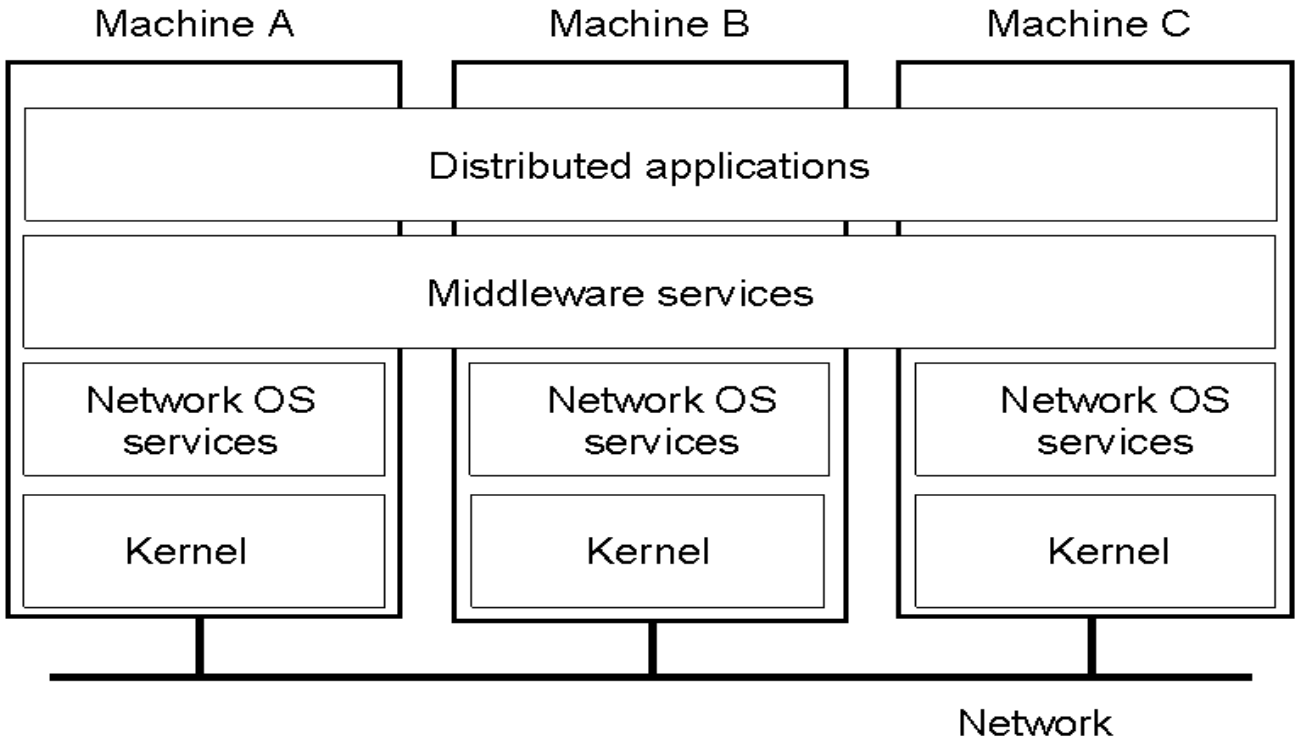
(b)



(c)

1.5 Software support

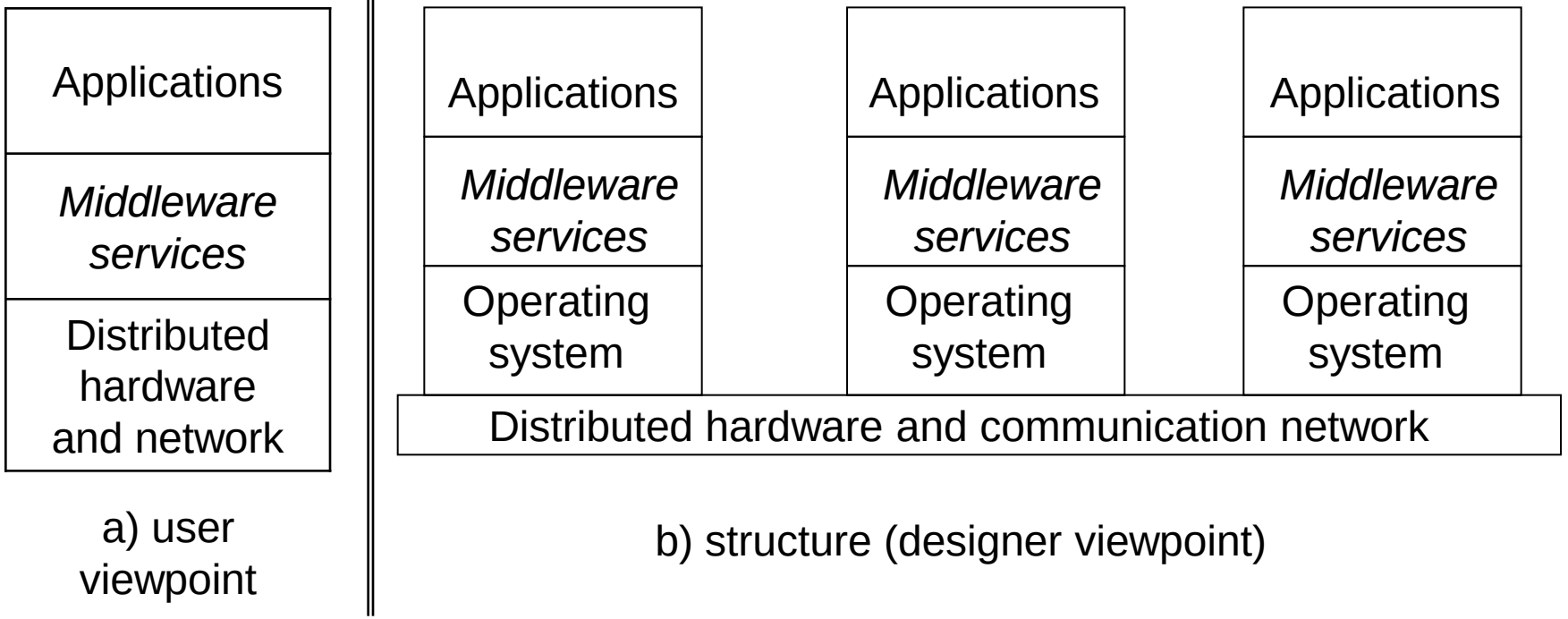
- Middleware



Comparison of DOS, NOS and Middleware

Item	Distributed OS		Network OS	Middleware-based OS
	Multiproc.	**Multicomp.**		
Degree of transparency	Very High	High	Low	High
Same OS on all nodes	Yes	Yes	No	No
Number of copies of OS	1	N	N	N
Basis for communication	Shared memory	Messages	Files	Model specific
Resource management	Global, central	Global, distributed	Per node	Per node
Scalability	No	Moderately	Yes	Varies
Openness	Closed	Closed	Open	Open

1.6 Structure of a distributed system



• *Middleware* services: RPC/RMI support, group-communication support, time synchronization, consistency (replication), name services, security services…